

Statistical and Computational Models

Lecture 8: Interpreting binary outcome models

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The fitted logit

At the end of the last slide deck we fit this model:

```
logit <- glm(y ~ oil + lgdp, data = countries, family = binomial,  
            control = glm.control(epsilon = 1e-10))  
round(summary(logit)$coefficients[, 1:2], 3)
```

	Estimate	Std. Error
(Intercept)	-15.329	2.343
oil	-1.560	0.429
lgdp	1.851	0.273

Three questions remain:

1. How do we characterize uncertainty?
2. How should we interpret the coefficients?
3. How is this statistical learning?

Roadmap

Uncertainty

Interpretation

Classification

The curvature of the log-likelihood

Recall that a spikier ℓ pins $\hat{\beta}$ down more tightly, so uncertainty is given by its curvature.

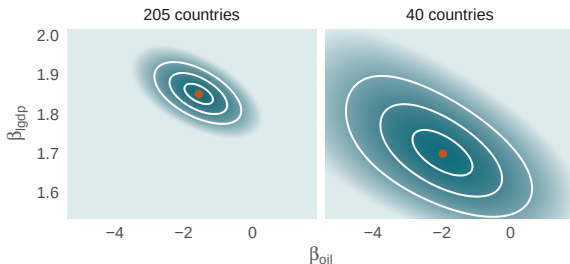
Curvature is the **Hessian** – the $p \times p$ matrix of second derivatives, which tells us the **rate of change of the slope** of ℓ .

$$H(\beta) = - \sum_{i=1}^n \pi_i(1 - \pi_i) \mathbf{x}_i \mathbf{x}_i^\top, \quad -H(\beta) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$$

$\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$: fitted probabilities on the diagonal, with zeroes everywhere else.

Every weight $\pi_i(1 - \pi_i)$ on the diagonal of $\mathbf{W}$ is positive, so $-H$ is positive definite and ℓ is concave – its zero is the MLE.

More information, less variance



Visualise ℓ for different choices of β . The darker the colour, the closer to the **peak** shown in red.

The contour lines show ℓ for a common fall from the peak – the tighter they are the spikier ℓ . How far can β move before the fit gets worse by a set amount? With a flatter ℓ , further – $-H$ is smaller, so $\left[-H(\hat{\beta})\right]^{-1}$ larger.

From curvature to standard errors

We will omit the proof, but when n is large, the sampling distribution of $\hat{\beta}$ itself is approximately normal:

$$\hat{\beta} \stackrel{\text{approx.}}{\sim} N\left(\beta, \left[-H(\hat{\beta})\right]^{-1}\right)$$

$-H(\hat{\beta})$ is the **observed information**: recall that at the peak ℓ is concave, so H is negative, and the minus makes it positive.

$$\widehat{\text{Var}}(\hat{\beta}) = \left[\mathbf{X}^T \mathbf{W} \mathbf{X}\right]^{-1}$$

The variance of each $\hat{\beta}_j$ down the diagonal, their covariances off it (think about the tilting rings in the previous slide). The standard errors are the square roots of that diagonal.

`summary()` reports these for any model fit by maximum likelihood. You are assuming the model is **true**, and n large.

Standard errors in R

```
X    <- model.matrix(logit)
p    <- fitted(logit)
w    <- p * (1 - p)
V_eh <- solve(t(X) %*% diag(w) %*% X)
rbind(hand = sqrt(diag(V_eh)), glm = summary(logit)$coefficients[, 2])
```

	(Intercept)	oil	lgdp
hand	2.342988	0.4289639	0.2727491
glm	2.342988	0.4289639	0.2727491

Note: we must invert (`solve()`) the whole matrix in the last line.

General advice: if you doubt the logit form, `{sandwich}::vcovHC()`.

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What is β in logit?

Consider a single logit β_j inside our vector β :

$$\beta_j = \underbrace{\frac{\partial}{\partial x_{ij}}}_{\text{change in } x_{ij}} \log \underbrace{\frac{\pi_i}{1 - \pi_i}}_{\text{odds}}$$

Read: How the **log** of the **odds** of $Y = 1$ changes for a one-unit change in x_{ij} only, holding all other $x_{i,-j}$ fixed.

The **odds**: how likely $Y = 1$ is against how likely $Y = 0$ is. When $\pi = 0.5$ then the odds are? **1** – the relative probabilities of each event are even.

Probability, odds, and log odds

Recall that probability is constrained by $[0, 1]$.

Odds are **relative probabilities**, so they are constrained instead by $[0, \infty]$.

The **log odds** run $[-\infty, \infty]$, with $\log(\frac{0.5}{1-0.5}) = 0$.

π	Odds	Log odds
0.1	0.11	-2.2
0.5	1.00	0.0
0.9	9.00	2.2

- Probability greater than 0.5 means $Y = 1$ is more likely than $Y = 0$
- Odds greater than 1 means the same
- Log odds greater than 0 means the same

A one-unit change

Consider a binary variable (`oil` in our case):

$$\text{oil} = 1 : \quad \log \text{odds}_1 = \beta_0 + \beta_{\text{lgdp}} \text{lgdp} + \beta_{\text{oil}}$$

$$\text{oil} = 0 : \quad \log \text{odds}_0 = \beta_0 + \beta_{\text{lgdp}} \text{lgdp}$$

$$\log \text{odds}_1 - \log \text{odds}_0 = \log \frac{\text{odds}_1}{\text{odds}_0} = \beta_{\text{oil}}$$

Read: β_{oil} is the **log** of the **odds ratio** – the ratio of the odds when `oil` = 1 to the odds when `oil` = 0.

If we exponentiate β_{oil} we recover the **odds ratio**:

$$e^{\beta_{\text{oil}}} = \frac{\text{odds}_1}{\text{odds}_0}$$

Note that this gives us the same answer as taking the derivative because the log odds are **linear** in $\mathbf{x}$. This is not true for other quantities.

Interpreting $\hat{\beta}_j$

To this point we have been discussing β , the **estimand in our parametric model**. In reality we interpret $\hat{\beta}$, an **estimate** we recover given our sample.

Some guidance for interpreting a single $\hat{\beta}_j$:

1. Read the **sign**: oil lowers the odds of democracy, $\ln gdp$ raises them.
2. Exponentiate $\hat{\beta}_j$ to recover the **odds ratio**, $e^{\hat{\beta}_j}$: the multiplier on the odds for a one-unit change in x_j . An odds ratio greater than 1 means a positive change, an odds ratio lower than 1 a negative, and 1 means no change.
3. For a change of size c , the multiplier is simply $e^{c\hat{\beta}_j}$. Doubling GDP per capita is $c = \log 2$, so 3.6.

Note: this is **not general** – these rules work because of the **logit link**.

Other GLMs will require **different** step 2s and 3s, and even step 1 is likewise not guaranteed!

Interpreting odds ratios

π	π with oil	Odds ratio	Change (pp)
0.1	0.02	0.21	8
0.5	0.17	0.21	33
0.9	0.65	0.21	25

Read: for a given lgdp , switching oil on multiplies the **odds** of democracy by $e^{\hat{\beta}_{\text{oil}}} = 0.21$.

The same odds ratio has a **different substantive interpretation** depending on π . A country at $\pi_i = 0.10$ loses 8 percentage points, while one at 0.50 loses 33 percentage points.

The size of the change depends on where π_i starts, which in turn depends on all of $\mathbf{x}_i$. But we do not know π_i – we have to **estimate** it!

Predicted probabilities

Define the **predicted probability** when $\mathbf{x}_i = \mathbf{x}$:

$$\hat{\pi}(\mathbf{x}) = \text{logit}^{-1}(\mathbf{x}^\top \hat{\beta})$$

Read: for a given realization of the covariates $\mathbf{x}$, what is the estimated probability of $Y = 1$ given our estimated $\hat{\beta}$?

Recovering these is easy – we multiply the model matrix by the estimated $\hat{\beta}$, wrapped in the inverse link function:

y	Log GDP	Oil	Fitted	True
1	9.77	1	0.768	0.809
1	9.23	0	0.852	0.866
1	8.32	0	0.519	0.512
0	6.49	1	0.008	0.006

But how meaningful is a vector of predicted probabilities alone?

Contrasts in predicted probabilities

Consider instead the following **contrast**: calculate $\hat{\pi}$ at one value of x_j , and subtract $\hat{\pi}$ for another.

$$\hat{\tau}_j(\mathbf{x}) = \hat{\pi}(x_j = \mathbf{a}, \mathbf{x}_{-j}) - \hat{\pi}(x_j = \mathbf{b}, \mathbf{x}_{-j})$$

For a binary variable (e.g. `oil`) we naturally evaluate $\mathbf{a} = 1$ and $\mathbf{b} = 0$.

For continuous variables (e.g. `lgdp`) we take the derivative, and the contrast at the limit becomes a slope:

$$\frac{\partial \hat{\pi}(\mathbf{x})}{\partial x_j} = \hat{\beta}_j \hat{\pi}(\mathbf{x}) (1 - \hat{\pi}(\mathbf{x}))$$

These are called **partial effects**, and hinge on two choices: which contrast, and at which values of $\mathbf{x}$?

Step 1: choosing the contrast

In our example, a one unit increase in $\ln \text{gdp}$ multiplies GDP per capita by 2.72.

Shrink the contrast to nothing and you have the derivative. Alternatively:

Contrast	Effect (pp)
One unit of $\ln \text{gdp}$	21.7
Doubling GDP per capita	16.0
One sd of $\ln \text{gdp}$	40.4
Lower to upper quartile	49.7

Report quantities that are meaningful: e.g. “**doubling GDP per capita** raises $\text{Pr}(\text{democracy})$ by 16 percentage points, on average.”

Step 2: choosing the values of $\mathbf{x}$

Recall we want to understand how moving from $\text{oil} = 0$ to $\text{oil} = 1$ changes $\hat{\pi}$ for a fixed lgdp . Which lgdp ?

PARTIAL EFFECT AT THE AVERAGE (PEA)

$$\begin{aligned}\hat{\tau}_{\text{PEA}} &= \hat{\pi}(1, \overline{\text{lgdp}}) - \hat{\pi}(0, \overline{\text{lgdp}}) \\ &= -0.366\end{aligned}$$

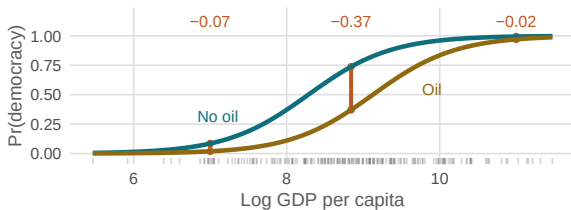
Read: the change in $\hat{\pi}$ for an 'average' country. For lgdp , that country has $\text{oil} = 0.28$.

AVERAGE PARTIAL EFFECT (APE)

$$\begin{aligned}\hat{\tau}_i &= \hat{\pi}(1, \text{lgdp}_i) - \hat{\pi}(0, \text{lgdp}_i) \\ \hat{\tau}_{\text{APE}} &= \frac{1}{n} \sum_{i=1}^n \hat{\tau}_i = -0.229\end{aligned}$$

Read: the change in $\hat{\pi}$ for each country at its own $\mathbf{x}_i$, then averaged.

Average case or observed values



The effect for the 'average' country need not be the **average of the effects**. oil shifts the log odds by the same $\hat{\beta}_{oil}$ at every 1 lgdp ; but how this aggregates depends on the distribution of the data.

$\hat{\tau}_{PEA}$ is the middle segment. The APE generates a segment at every tick (every unit) instead, then averages that.

Hanmer and Kalkan (2013) recommend the **observed-value** approach: the target is the average effect in the population.

Calculating the PEA by hand

Create two quantities, with `lgdp` set to the mean of all countries, once with `oil` off and once on.

This is just two predictions and one difference:

```
at_mean <- data.frame(lgdp = mean(countries$lgdp), oil = 0:1)
pea <- unname(diff(predict(logit, at_mean, type = "response")))
round(pea, 3)
```

```
[1] -0.366
```

Calculating the APE by hand

Create two **counterfactual copies** of the data: one with `oil` set to 0 for everyone, one with it set to 1, but everything else unchanged.

Oil	$\hat{\pi}$	If no oil	If oil	Difference
1	0.768	0.940	0.768	-0.172
0	0.852	0.852	0.548	-0.304
0	0.519	0.519	0.185	-0.334

Each country's estimated $\hat{\pi}$ is one of its two counterfactuals.

```
cf0 <- cf1 <- countries
cf0$oil <- 0; cf1$oil <- 1
ape <- mean(predict(logit, cf1, type = "response") -
             predict(logit, cf0, type = "response"))
round(ape, 3)
```

```
[1] -0.229
```

Quantities of interest in R

These days we can use `{marginaleffects}` (Arel-Bundock, *Model to Meaning*) to make the calculation of quantities of interest more straightforward:

```
library(marginaleffects)
avg_comparisons(logit, variables = "oil")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
-0.229	0.0599	-3.82	<0.001	12.8	-0.346	-0.111

Term: oil

Type: response

Comparison: 1 - 0

This gives us the APE, recovering the same **-0.229** but with a standard error (which can be made robust) and a corresponding interval.

Using {modelsummary}

{modelsummary} takes the fit and the quantities in one call:

```
library(modelsummary)
qoi <- avg_comparisons(logit, variables = list(oil = 0:1, lgdp = log(2)))
modelsummary(list("Logit coef." = logit, "Avg. effect" = qoi),
  statistic = "std.error", gof_map = NA,
  coef_omit = "Intercept")
```

	Logit coef.	Avg. effect
oil	-1.560 (0.429)	-0.229 (0.060)
lgdp	1.851 (0.273)	0.160 (0.010)

Lead with interpretability – e.g. the right-hand column.

Uncertainty for a quantity of interest

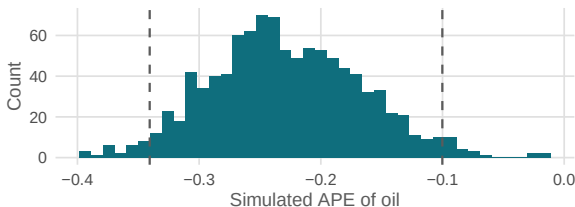
`summary()` prints standard errors for $\hat{\beta}$. We want one for a function of it, $T(\hat{\beta})$ – a predicted probability, a PEA, an APE.

$\hat{\beta}$ is approximately normal, so **simulate** from $N(\hat{\beta}, \widehat{\text{Var}}(\hat{\beta}))$: variance in the draws of β transfers to $T(\beta)$.

1. Draw $\beta^{(k)}, k = 1, \dots, R$.
2. Plug each into T : $T(\beta^{(k)})$.
3. se: their sd. 95% interval: their 2.5 and 97.5 percentiles.

King, Tomz and Wittenberg (2000); `inferences()` in `{marginaleffects}`.

Uncertainty for a quantity of interest



T is the APE of oil, $R = 1000$ draws. Dashed: the 2.5 and 97.5 percentiles.

$\hat{se} = 0.062$, 95% interval $[-0.34, -0.10]$.

The **delta method** (a Taylor expansion of T around $\hat{\beta}$) is the analytic route, and the package's default: $\hat{se} = 0.060$.

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0-1 loss

Suppose we need a **predicted outcome** $\hat{y}_i$ at a threshold q (customarily 0.5), and score the misses with the **0-1 loss**:

$$\text{Err}_i = \mathbf{1}\{y_i \neq \hat{y}_i\}, \quad \hat{y}_i = \mathbf{1}\{\hat{\pi}_i \geq q\}$$

	$y_i = 0$	$y_i = 1$
$\hat{y}_i = 0$	60 true negatives	18 false negatives
$\hat{y}_i = 1$	24 false positives	103 true positives

Percent correctly predicted is the diagonal over n :
 $(60 + 103)/205 = 79.5\%$.

Logit is actually just a classifier!

References

- Arel-Bundock, V. (2026) *Model to Meaning*. CRC Press – chs. 5–7. marginaleffects.com.
- Hanmer, M. J. & Kalkan, K. O. (2013) "Behind the Curve: Clarifying the Best Approach to Calculating Predicted Probabilities and Marginal Effects from Limited Dependent Variable Models." *American Journal of Political Science* 57(1): 263–277.